Reaction curation app - Guide with Example

1. Progress overview and reaction selection
2. Inspect reaction and atom balance
3. Use molecule editor to curate reaction
4. Inside the molecule editor
5. Describe curation and metadata collection

More information:

- Patent viewer guide

1. Progress overview and reaction selection

Logged in as: gabrielvogel

Logout

Chemical Reaction Curation Tool

If you use this tool for the first time, please refer to the guide:

Guide

Your Progress

Assigned	Completed	Skipped	Pending
51	3	0	48

Your assigned reactions. Ideally all of them should be completed, not skipped.

› Reactions Overview

Select Reaction

Select the a specific reaction or navigate using the buttons

Choose reaction

Reaction 4

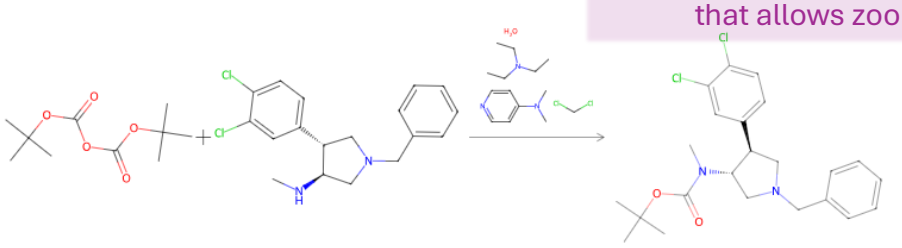
Previous

Next

Visualization of reaction for inspection

Original Reaction

Enlarge (opens a small popup that allows zooming) →



Original Reaction

Browse Reactant 1 in PubChem

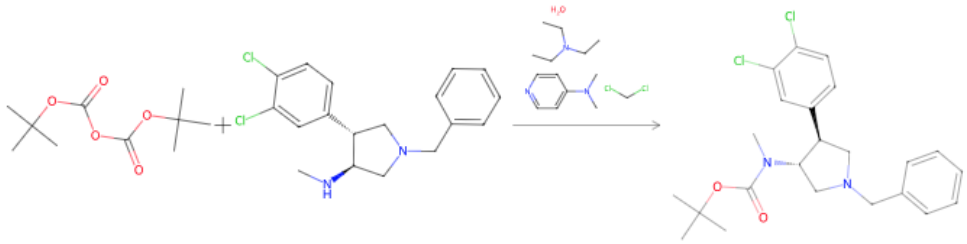
Browse Reactant 2 in PubChem

Browse Product 1 in PubChem

› View SMILES

2. Inspect the reaction and the Atom Balance

Original Reaction



Original Reaction

Links to browse molecules in PubChem

Browse Reactant 1 in PubChem Browse Reactant 2 in PubChem Browse Product 1 in PubChem

View SMILES

SMILES is a text representation of molecules – allows copying parts.
Pattern: reactant.reactant>reagents>product

```
CC(C)(C)OC(=O)OC(=O)OC(C)(C)C.CN[C@@H]1CN(Cc2ccccc2)C[C@H]1c1ccc(Cl)c(Cl)c1>CCN(CC)CC.CN(C)c1ccncc1.Cl
```

Balance Check:

Net left missing atoms	Net right missing atoms	Charge imbalance
None	C:5 H:10 O:3	None

What does the balance mean?

Explanation how to interpret the balance

The balance deficit shows what is **at least** needed to balance the reaction.

- **Left** lists atoms that would need to be **added to the left side** to match the right.
- **Right** lists atoms that would need to be **added to the right side** to match the left.
- **Charge** is the net charge difference between both sides (0 means charge-balanced).

IMPORTANT: The deficit does not necessarily reflect all actual missing atoms per side.

If both Left and Right are empty (OK) and Charge is 0, the reaction is balanced.

Here: The balance means that there are 5 carbon atoms, 10 hydrogen atoms and 3 oxygen atoms missing on the product side of the equation.
The reagents don't count towards the atom balance
What could this mean chemically?

- We can see that this is a typical Boc protection of the secondary amine.
- The reported product is a carbamate (Boc-protected amine), but the byproduct(s) are missing → tert-butyl hydrogen carbonate (typically decomposing to tert-Butanol and carbon dioxide) are missing.

→ This means we need to add those molecules to the product side.

3. Use Molecule Editor to curate the reaction

Molecule Editor

In this section you can move, modify, add, or remove molecules for this reaction in order to make it a complete and balanced reaction. To draw molecules and export the SMILES string, you can use e.g., Sketcher.

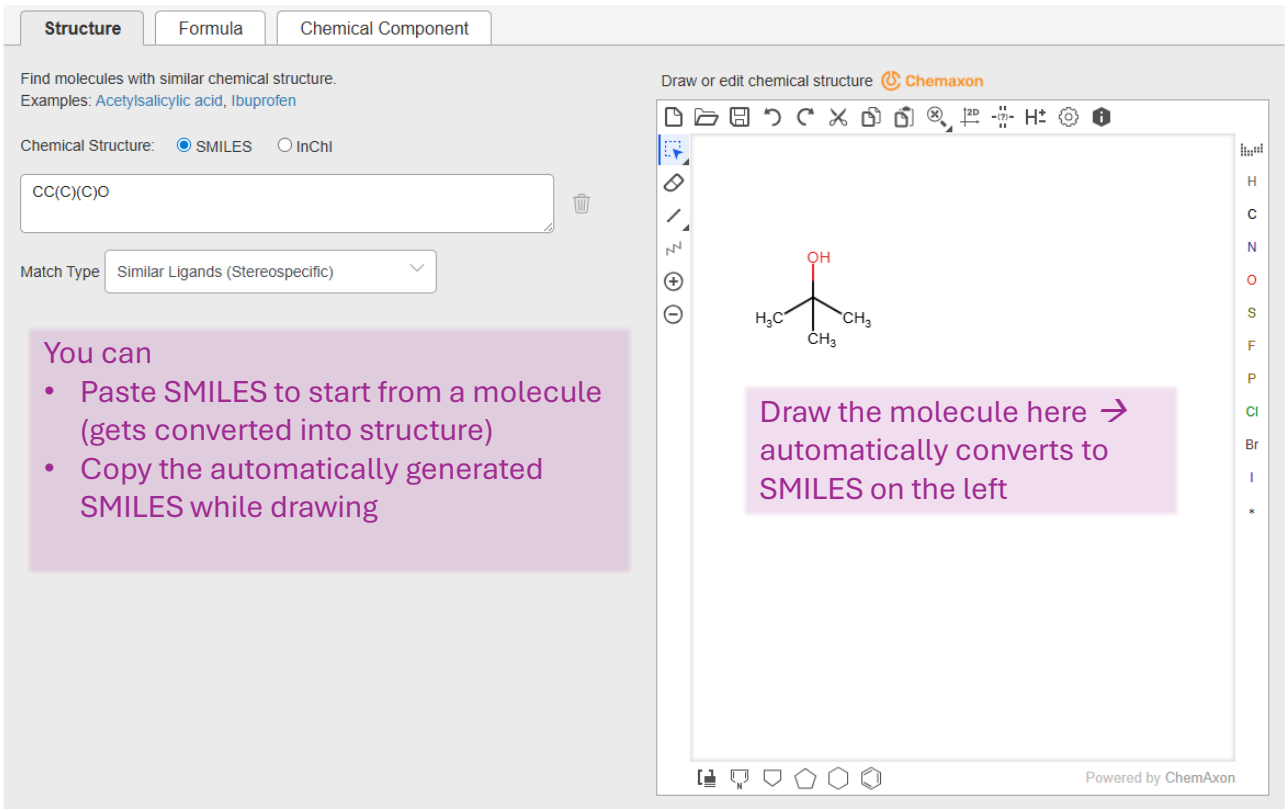
☐ Modify original molecules 1.

+ Add molecule 2.

Recompute Balance & Visualize 3.

1. The checkbox let's you modify the original molecules
   - modify their SMILES representation to edit molecule
   - change the quantity (stoichiometry)
   - change the role (reactant, reagent, product)
   - Delete molecules
2. The add molecule button let's you create new molecules
   - Add a new reactant, product or reagent
   - Needs to be added inserted as SMILES string (you can draw in the linked Sketcher and export the SMILES if you don't know how to write SMILES), see example below
3. Recompute Balance & Visualize
   - Recomputes the balance of your edited reaction and visualizes it

Example to get SMILES for molecule via Sketcher



4. Inside the Molecule Editor

Molecule Editor

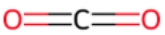
In this section you can move, modify, add, or remove molecules for this reaction in order to make it a complete and balanced reaction. To draw molecules and export the SMILES string, you can use e.g., Sketcher. ← Use to draw molecules and get SMILES

☐ Modify original molecules ← Option to modify/move the recorded molecules

CC(C)(C)O 1. ✓ 2. OK | 1 | 3. Products | 🗑



O=C=O ✓ OK | 1 | Products | 🗑

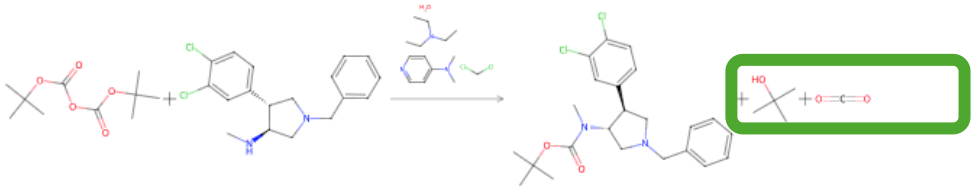


+ Add molecule

4. Recompute Balance & Visualize

Balanced after modifications — Reaction is balanced = Success (but be certain about correctness)

Updated Reaction Preview



› View Original reaction as reference

Steps 1-4:
1. Insert SMILES (either handwrite or export from drawer/Pubchem)
2. Validate SMILES
3. Choose Reactant/Reagent/Product role + Quantity
4. Recompute the balance

5. Describe curation and record metadata

Describe Your Curation

How certain are you with the curated reaction? (1 = low, 5 = high)*

1 | 2 | 3 | 4 | 5

What did you change? *

Added missing product(s)

Explanation *

This is a typical Boc protection of the secondary amine.
The reported product is a carbamate (Boc-protected amine), but the byproducts tert-Butanol and carbon dioxide are missing.

Did you look up the reaction in the patent?*

- Select an option...
- Yes
- No

It's not mandatory, but can be helpful, if the reaction does not seem to make sense!

Select the reaction mechanism class possible:*

Protections

Does it fit in a predefined class? Or specify your own classification

Did you use any additional sources (e.g., literature, databases) to inform your curation? If so, please specify (links):

Optional - E.g., journal articles, chemical databases, etc.

Please mention if you found a significant source that helped you solve this reaction curation

Did you use any AI tools (e.g., ChatGPT, Copilot) to assist in your curation?*

Please do not use AI if you know yourself

- Select an option...
- Yes
- No

If you used AI as help, make sure to double check the retrieved information with alternative sources and your chemistry knowledge

- [x] I critically evaluated and verified all AI-generated suggestions to ensure accuracy and reliability.*

Save your curated reaction

Save and Next ← Saves and locks the reaction

Reset This Reaction ← Option to reset all the modifications and start over from scratch

More information

- Alternative solution for this example
- Still imbalanced view
- How to use patent browser
- Skip reaction

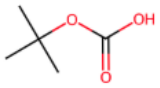
Alternative solution for this example

Molecule Editor

In this section you can move, modify, add, or remove molecules for this reaction in order to make it a complete and balanced reaction. To draw molecules and export the SMILES string, you can use e.g., Sketcher.

☐ Modify original molecules

CC(C)(C)OC(=O)O | ✓ | OK | 1 | Products

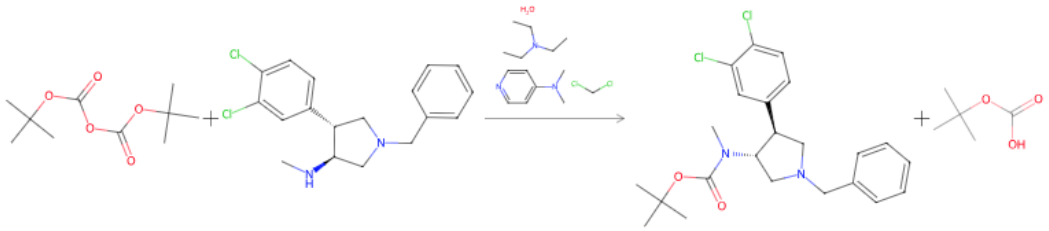


+ Add molecule

Recompute Balance & Visualize

Balanced after modifications

Updated Reaction Preview



› View Original reaction as reference

Reaction still imbalanced view

Molecule Editor

In this section you can move, modify, add, or remove molecules for this reaction in order to make it a complete and balanced reaction. To draw molecules and export the SMILES string, you can use e.g., Sketcher.

☐ Modify original molecules

CC(C)(C)O | ✓ | OK | 1 | Products | 🗑



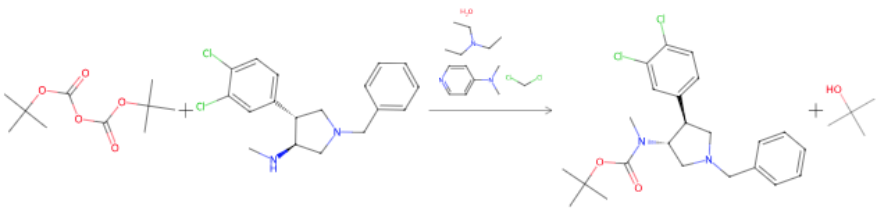
+ Add molecule

Recompute Balance & Visualize

Still imbalanced

Net left missing atoms	Net right missing atoms	Charge imbalance
None	C:1 O:2	None

Updated Reaction Preview



› View Original reaction as reference

After adding tert-butanol, there are still one carbon and two oxygen atoms missing on the right side (Carbon dioxide).

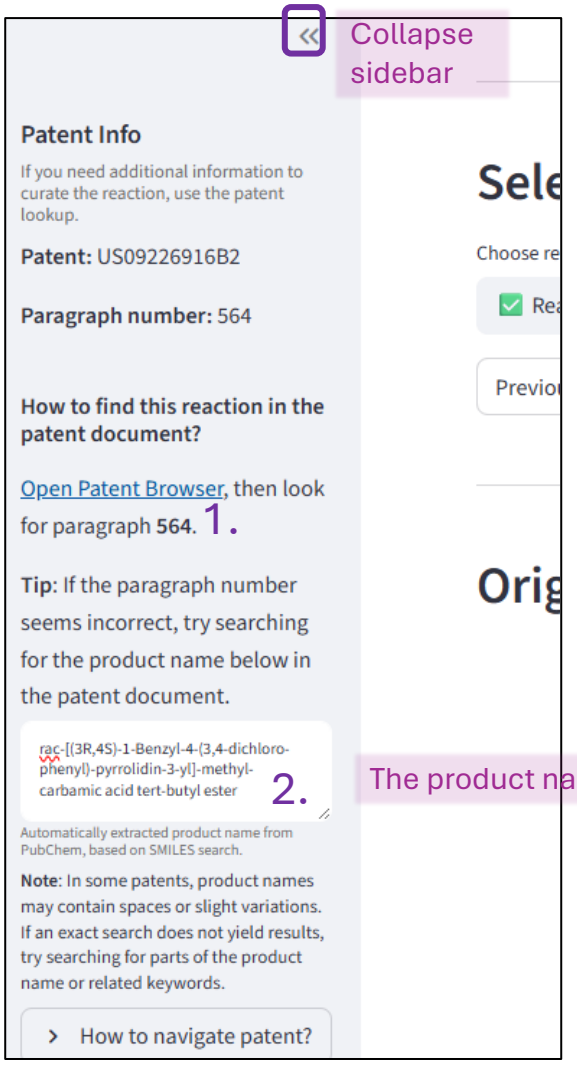
Patent viewer

Expand sidebar with patent info



Patent viewer

Collapse sidebar



Some notes on the patent browsing

- The patents are very long and contain multiple reactions most times
- It can be hard to find the exact reaction

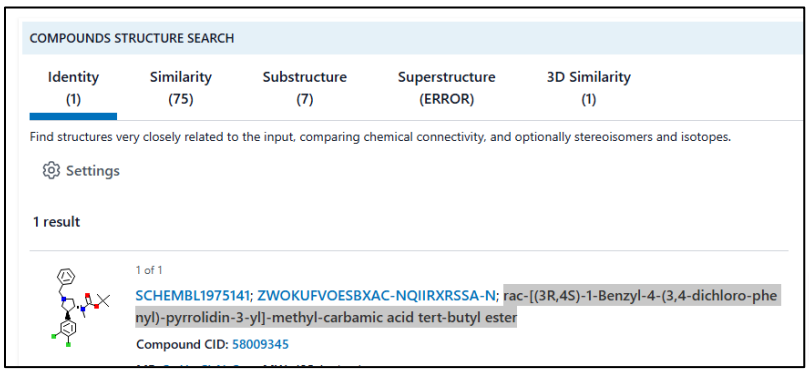
Try the following:

1. If available, search for the paragraph number
2. Alternatively, copy and search for the automatically extracted product name from PubChem (see next page)

The product name we extract from PubChem for you

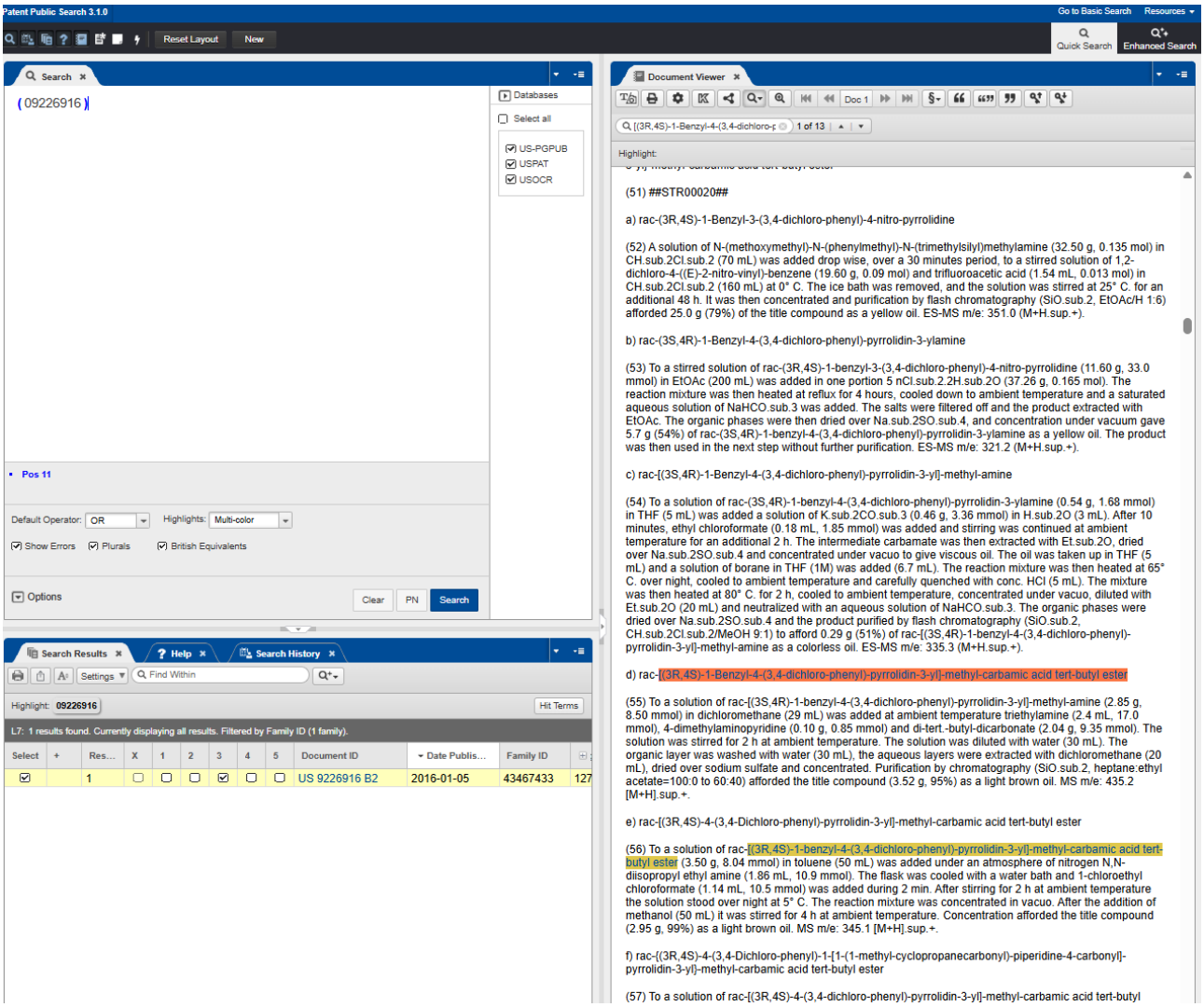
Explanations how to use the patent browser

PubChem



The product name we extract from PubChem for you

Patent viewer



- In this case we see that the actual paragraph is 55 (and not 564 as recorded).
- The experimental procedure matches the extracted reaction
    - There is more information on the purification, but no mention of the byproducts → need to be inferred by us chemists via knowledge